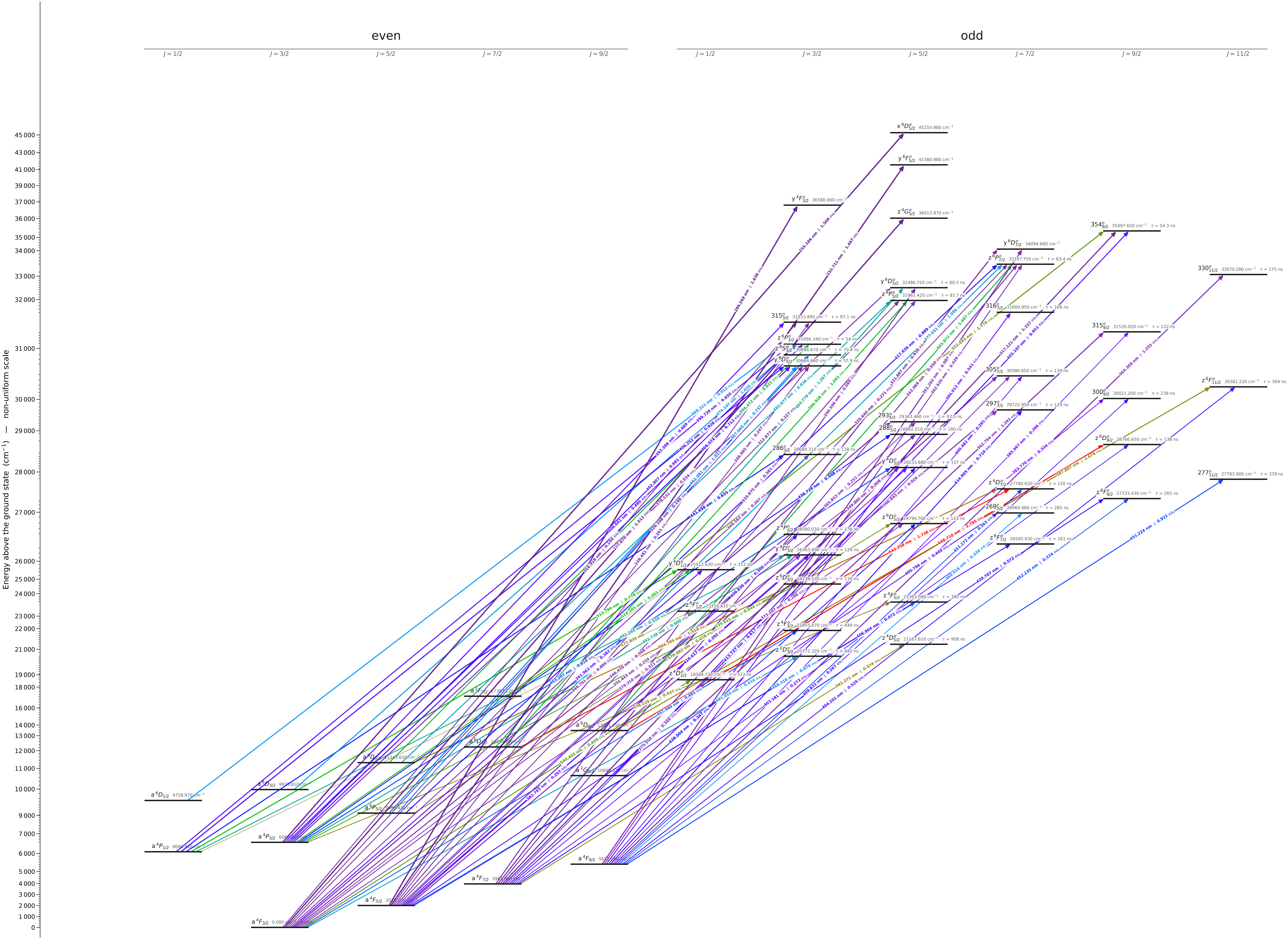
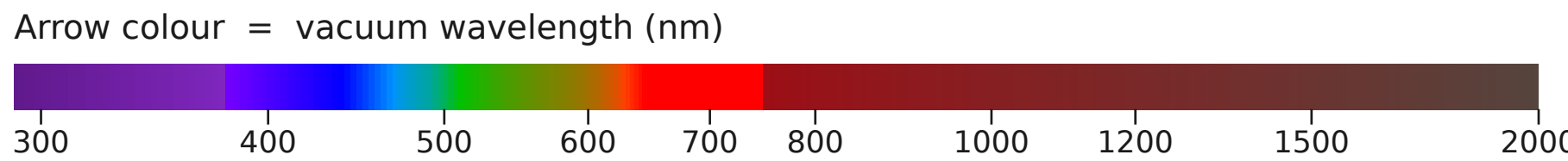


Tantalum

Ta (Ta I). Strongest classified transitions below 2 μm (NIST): 55 levels, 90 lines.

Ta+ ionisation limit 60891.40 cm⁻¹ = 7.54957 eV (164.227 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J || e r || J' \rangle$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle J e r J' \rangle$ ($e a_0$)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$a^4P_{5/2} - z^4D_{5/2}^o$	570.0822	569.9241	525.8758	0.478	4.167e+05	1.408	4.6	hyperfine-structure spectroscopy line
$a^4P_{1/2} - z^4F_{1/2}^o$	577.8346	577.6744	518.8205	-	-	-	-	hyperfine-structure spectroscopy line
$a^4P_{3/2} - z^4F_{3/2}^o$	578.2292	578.0689	518.4665	0.437	3.333e+05	0.4081	13	hyperfine-structure spectroscopy line
$a^6D_{9/2} - z^6F_{11/2}^o$	587.8974	587.7345	509.9401	1.676	2.333e+06	0.5235	72	hyperfine-structure spectroscopy line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: Measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁸¹Ta hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$a^4P_{1/2}$	879(10)	-	F=3: -1977.75 F=4: +1538.25
$a^4P_{3/2}$	378(10)	-1349(20)	F=2: -3274.18 F=3: -1176.61 F=4: +720.82 F=5: +1647.25
$a^4P_{5/2}$	251(10)	1670(20)	F=1: -1929.11 F=2: -1904.25 F=3: -1688.04 F=4: -1065.75 F=5: +248.89 F=6: +2613.75
$a^6D_{9/2}$	1089(10)	-	F=1: -20963.25 F=2: -18785.25 F=3: -15518.25 F=4: -11162.25 F=5: -5717.25 F=6: +816.75 F=7: +8439.75 F=8: +17151.75
$z^6F_{1/2}^o$	-1771(10)	-	F=3: +3984.75 F=4: -3099.25
$z^4F_{5/2}^o$	480(10)	-19(20)	F=1: -5410.18 F=2: -4444.75 F=3: -2998.64 F=4: -1074.30 F=5: +1325.02 F=6: +4195.25
$z^6D_{5/2}^o$	1033(10)	688(20)	F=1: -11252.68 F=2: -9383.25 F=3: -6505.39 F=4: -2530.65 F=5: +2658.92 F=6: +9210.75
$z^6F_{3/2}^o$	669(10)	60(20)	F=2: -15194.93 F=3: -13197.75 F=4: -10532.66 F=5: -7197.79 F=6: -3190.80 F=7: +1491.11 F=8: +6851.21 F=9: +12893.25

Measured A, B constants; sources in the data file.